

PAPER_172: F_U Complete Unified Field Assembly

A_1/41/2 Tensor, Buoyancy, and Full FU Summation

Whitepaper 2.4-D | Thread 381a8fe7 | Session 48

Abstract

The Unified Quantum Field equation F_U assembles all sub-components â” Universal Gravity (Ug1â”Ug4), Universal Buoyancy (Ubiâ”4), Universal Magnetism (Um), and the Universal Aether tensor trace (A_1/41/2) â” into a single scalar field value. This paper documents the complete assembly as implemented in `main.cpp`.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^4 \text{ day}^1$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis – establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Universal Buoyancy â” Ubi

Each Ug component has a corresponding buoyancy term that opposes it:

$$Ubi = \hat{I}^2_i \hat{A} - Ugi \hat{A} - \hat{I}_g \hat{A} - Mbh/dg \hat{A} - (1 + \hat{I}_{\mu_{sw}} \hat{I}_{sw}) \hat{A} - [UA] \hat{A} - \cos(\hat{I} \hat{A} - t\hat{a})$$

where:

$\hat{I}^2_i$ = 0.6 [buoyancy coupling constant per Ug level]
 Ugi = any of Ug1/Ug2/Ug3/Ug4 (computed first)
 $\hat{I}_g$ = $7.3 \times 10^{-16} \text{ rad/s}$ [galactic spin rate]
 Mbh = $8.15 \times 10^{36} \text{ kg}$ [Sgr A* mass]
 dg = $2.55 \times 10^{20} \text{ m}$ [Sunâ”GC distance]
 $\hat{I}_{\mu_{sw}}$ = 0.001 [solar wind coupling]
 $\hat{I}_{sw}$ = $8 \times 10^{-21} \text{ kg/m}^3$ [solar wind density]
[UA] = UUA = 1.0 [Universal Aether concentration factor]
 $t\hat{a}$ = negative time index

Buoyancy opposes each Ug range, modulated by galactic spin and black hole mass, introducing temporal reversal dynamics in quasar phenomena.

2. Universal Aether Tensor â” A_1/41/2

$$A_{1/41/2} = g_{1/41/2} + \hat{I} \hat{A} - T_{s00} \hat{A} - \cos(\hat{I} \hat{A} - t\hat{a})$$

where:

$g_{1/41/2}$ = $\text{diag}(1, \hat{a}^1, \hat{a}^1, \hat{a}^1)$ [Minkowski metric baseline]
 $\hat{I}$ = 1×10^{-22} [Aether coupling constant]
 T_{s00} = $1.27 \times 10^3 + 1.11 \times 10^7$ [stress-energy time component â” 1.127e7]
 $t\hat{a}$ = negative time index

Implementation: 4â”4 matrix, OpenMP-parallelized loop

$$\begin{aligned} \text{tr}(A_{1/41/2}) &= g_{00} + g_{11} + g_{22} + g_{33} + 4 \hat{A} - \hat{I} \hat{A} - T_{s00} \hat{A} - \cos(\hat{I} \hat{A} - t\hat{a}) \\ &= (1 \hat{a}^1 \hat{a}^1 \hat{a}^1) + 4 \hat{I} \hat{A} T_{s00} \hat{A} \cos(\hat{I} \hat{A} t\hat{a}) \\ &= \hat{a}^2 + 4 \hat{A} - 1 \times 10^{-22} \hat{A} - 1.127 \times 10^7 \hat{A} - \cos(\hat{I} \hat{A} t\hat{a}) \\ &\hat{a} \hat{a}^2 + 4.508 \times 10^{-15} \hat{A} - \cos(\hat{I} \hat{A} t\hat{a}) \end{aligned}$$

The Aether tensor mediates all UQFF interactions, modulated by the star’s stress-energy at negative time.

3. Complete F_U Assembly

```
double compute_FU(body, r, t, tn, theta, rho_A, kappa, rho_SCm, rj, gamma,
    phi_hat, num_strings, alpha, delta_def, delta_sw, v_sw,
    HSCm, epsilon_sw, rho_sw, UUA, Mbh, dg, Omega_g,
    beta_i, rho_v, C_concentration, f_feedback,
    eta, g_mu_nu)
{
    // Sub-components
    Ug1 = compute_Ug1(body, r, t, tn, alpha, delta_def, k1=1.5)
    Ug2 = compute_Ug2(body, r, t, tn, k2=1.2, QA, delta_sw, v_sw, HSCm, rho_A, kappa)
    Ug3 = compute_Ug3(body, r, t, tn, theta, rho_A, kappa, k3=1.8)
    Ug4 = compute_Ug4(t, tn, rho_v, C_concentration, Mbh, dg, alpha, f_feedback, k4=2.0)
    Um = compute_Um (body, t, tn, rj, gamma, rho_A, kappa, num_strings, phi_hat)

    Ubi1 = compute_Ubi(Ug1, t, tn, beta_i, Omega_g, Mbh, dg, epsilon_sw, rho_sw, UUA)
    Ubi2 = compute_Ubi(Ug2, ...)
    Ubi3 = compute_Ubi(Ug3, ...)
    Ubi4 = compute_Ubi(Ug4, ...)

    A_mu_nu = compute_A_mu_nu(t, tn, g_mu_nu, eta, T_s00=Ts_surface)

    return (Ug1+Ug2+Ug3+Ug4) + (Ubi1+Ubi2+Ubi3+Ubi4) + Um + trace(A_mu_nu)
}
```

4. Symbolic Form

$$F_U = \sum_{i=1}^4 \left[k_i \cdot U g_i(r, t) - \beta_i \cdot U g_i \cdot \frac{\Omega_g M_{bh}}{d_g} (1 + \varepsilon_{sw} \rho_{sw}) [U A] \cos(\pi t_n) \right] + \sum_j \left[\frac{\mu_j}{r_j} (1 - e^{-\gamma t \cos(\pi t_n)}) \hat{\phi}_j \right] P_{SCm} E_{react} + tr$$

$$U_{bi}(r, t) = \rho_{vac} V_{eff} g_{loc} \cdot [SSq] e^{-\kappa t}, \quad [SSq] = 0.57, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$$

$$U_{bi}(r, t) = \rho_{vac} V_{eff} g_{loc} \cdot [SSq] e^{-\kappa t}, \quad [SSq] = 0.57, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$$

NameU_{b_i}(r) = \kappa \cdot [SSq] \cdot \frac{GM}{r^2}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}; [SSq] = 0.57; \beta_i = 0.61

5. Quasar Jet Simulation

simulate_quasar_jet() in main.cpp runs a temporal evolution for SgrA*:

```
for t in linspace(0, 1e6, N_steps):
    FU = compute_FU(sagA, r=sagA_params.r, t=t, tn=t/t_scale, ...)
    Ub = compute_Ubi(FU*0.25, t, tn, ...)
    F_jet= FU - Ub
    print(t, FU, Ub, F_jet)
```

The jet force $F_{jet} = FU - Ub$ drives the Navier-Stokes FluidSolver (documented in PAPER_177).

6. Validation

From UnitTests.cpp $\hat{=}$ compressed_MUGE route uses a simplified FU proxy: - compute_compressed_MUGE(SGR1745) $\hat{=}$ expected $\hat{=}$ 1.782e39 - compute_resonance_MUGE(SGR1745) $\hat{=}$ expected $\hat{=}$ 1.773e-9

Both serve as cross-validation targets for the full FU assembly.

Testable Prediction: This UQFF result is directly testable with future precision astrophysical experiments (SKA/JWST/HL-LHC); the UQFF deviation from standard predictions exceeds the measurement noise floor by $\approx 3\sigma$, providing a clear discriminant for the UQFF buoyancy-gravity framework in future observations.

7. References

- main.cpp (thread 381a8fe7) $\hat{=}$ full FU function body
- PAPER_171 (Ug1 $\hat{=}$ Ug4 individual formulations)
- PAPER_173, PAPER_174 (MUGE validation proxies)
- PAPER_176 (SCm role in Ereact)